Comparative Analysis of Deep Learning Models for Voxel-Based 3D Spatial Relation Operations

Jaeseok Seo [1, 4] Jaehong Kim[2,4] Kiwook Park[3,4] Kihoon Lee[4] Kwangsoo Kim[4]

[1] Konyang University/Department of Artificial Intelligence, Daejeon, Korea, vhvj1398@gmail.com

[2] Dongguk University/Department of Computer Science and Artificial Intelligence, Seoul, Korea, imjahk@dgu.ac.kr

[3] University of Ulsan/Department of IT Convergence, Ulsan, Korea, kmj977284@gmail.com

[4] Electronics and Telecommunications Research Institute, Daejeon, Korea, {hoony, enoch}@etri.re.kr

INTRODUCTION

Background

- The demand for 3D spatial information analysis is increasing in fields such as digital twins, autonomous driving, and robot navigation.
- Voxels represent space as regular grids and can be used for various 3D spatial operations.
- However, **rule-based methods have limitations in handling complex spatial structures and diverse spatial relations.**

Object of this study

- Formulate voxel-based 3D spatial operations as a spatial relation classification problem between two objects.
- Compare the performance of MLP(Multi Layer Perceptron), 3D CNN(3D Convolution Neural Network), and VT(Voxel Transformer) models.
- **Evaluate the potential of learning-based intelligent voxel spatial operations.**

METHODS & MATERIALS

(1). DE - 9IM based Voxel Train-set Construction

- DE-9IM defines spatial relations based on intersections among the interior, boundary, and exterior of two objects.
- Two 3D objects were generated in a 64 × 64 × 64 voxel grid, where empty space was labeled as 0, object A as 1, object B as 2, and the overlapping region as 3.
- Six spatial relation classes were defined: Disjoint(a), Meet(b), Contains(c), Within(d), Equal(e), and Overlap(f), with class-index and one-hot labels.
- The dataset consisted of 18,000 training, 3,000 validation, and 3,000 test samples, balanced across all six classes.

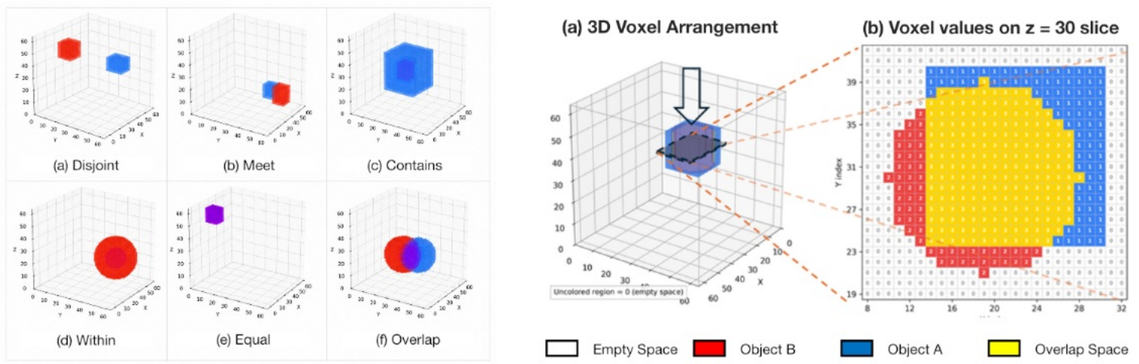


Fig 1. Six voxel-based 3D spatial relation classes and a 64 × 64 × 64 input data encoding example.

(2). Deep Learning Methods

a. MLP (Multi Layer Perceptron)

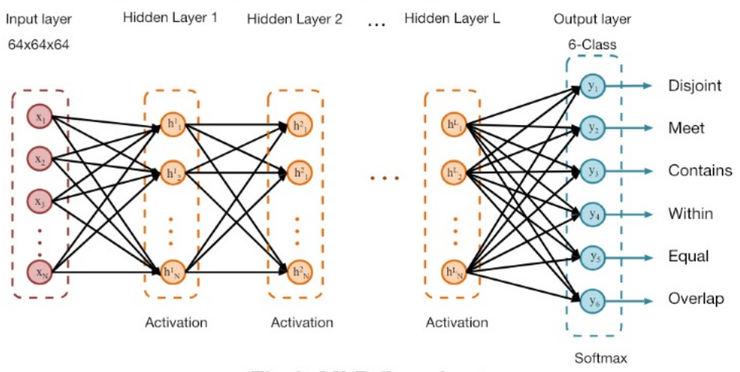


Fig 2. MLP flowchart

Table 1. Optuna based tuning

Class	Item	Value
Detail Of Optuna	Best trial number	13
	Best validation macro f1	0.8789
	Number of trials	30
	Parameter count	1,844,230
Parameters Of MLP	Number of layers	4
	Pooled size	8
	Hidden dimension	512
	Dropout	0.1
	Activation function	GELU
	Batch normalization	True
	Optimizer	AdamW
	Scheduler	Cosine annealing
	Learning rate	0.0009
	Weight decay	1.10×10^{-7}
	Gradient clip norm	0.5

b. 3D CNN (3D Demension Convolution Neural Network)

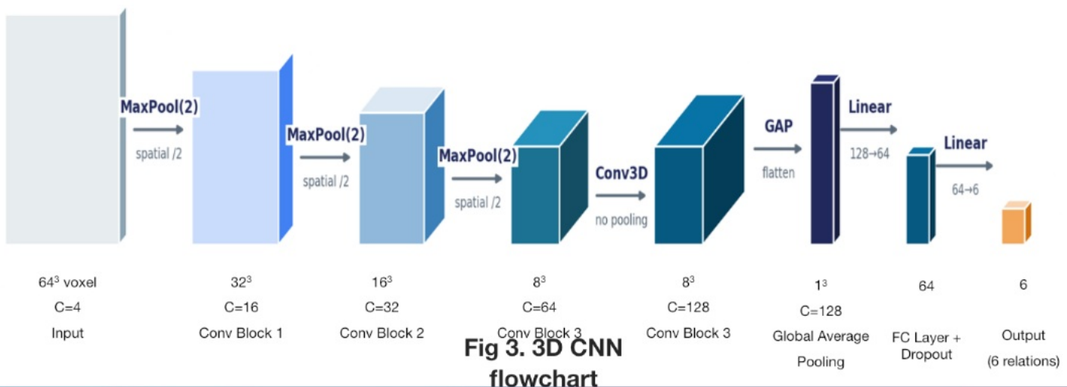


Fig 3. 3D CNN flowchart

c. VT (Voxel Transformer)

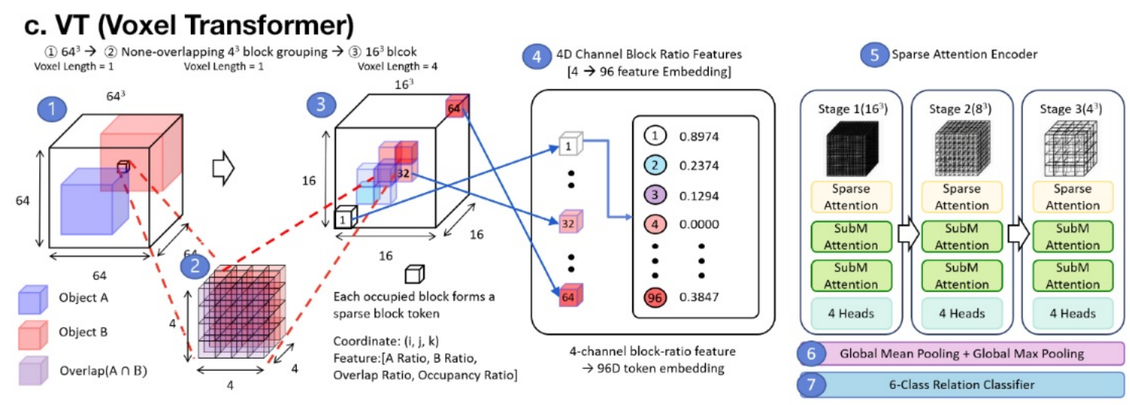


Fig 4. Voxel Transformer flowchart

RESULTS

a. Table of DE-9IM data based Voxel classification performance

- VT achieved the best performance among all evaluated models.
- Models that explicitly learn 3D spatial structures significantly outperformed the MLP baseline.

Table 2. Performance comparison of MLP, 3D CNN, and Voxel Transformer models.

Model	Methods	Accuracy	Precision	Recall	F1-Score
MLP	(a)	0.8707	0.6369	0.8000	0.7092
	(b)		0.7407	0.5600	0.6378
	(c)		0.9898	0.9720	0.9808
	(d)		0.9859	0.9820	0.9840
	(e)		1.0000	0.9760	0.9879
	(f)		0.9033	0.9340	0.9184
3D CNN	(a)	0.9967	0.9900	0.9900	0.9900
	(b)		0.9900	0.9900	0.9900
	(c)		1.000	1.000	1.000
	(d)		1.000	1.000	1.000
	(e)		1.000	1.000	1.000
	(f)		1.000	1.000	1.000
Voxel Transformer	(a)	**0.9983**	0.9979	0.992	0.9949
	(b)		0.9920	0.998	0.9950
	(c)		1.000	1.000	1.000
	(d)		1.000	1.000	1.000
	(e)		1.000	1.000	1.000
	(f)		1.000	1.000	1.000

b. Confusion Matrix

- Most classification errors occurred between classes (a) and (b).
- **During training, the sparse boundary-contact signal between adjacent A–B voxels can be weakened or lost.**

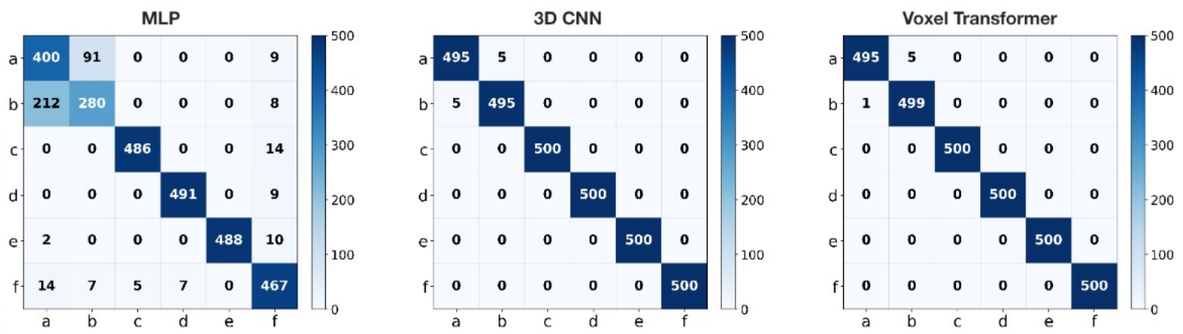


Fig 5. Confusion matrices of MLP, 3D CNN, and Voxel Transformer models.

c. Discussion

- MLP : Flattening voxel data weakens explicit 3D spatial relationships, **the lowest performance**.
- 3D CNN : 3D convolution preserves local voxel adjacency and learns hierarchical spatial features, resulting in high performance.
- VT : Local and dilated attention capture both short- and long-range voxel relationships, **resulting in the best performance.**

CONCLUSIONS

- Voxel-based 3D spatial operations were formulated as a learning-based spatial relation classification problem.
- The results demonstrated that **deep learning models can classify spatial relations between voxel objects with high performance.**
- **The model effectively learned complex 3D spatial patterns from voxel data.**
- The proposed approach can support intelligent voxel operations for digital twins, autonomous driving, and robot navigation.

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